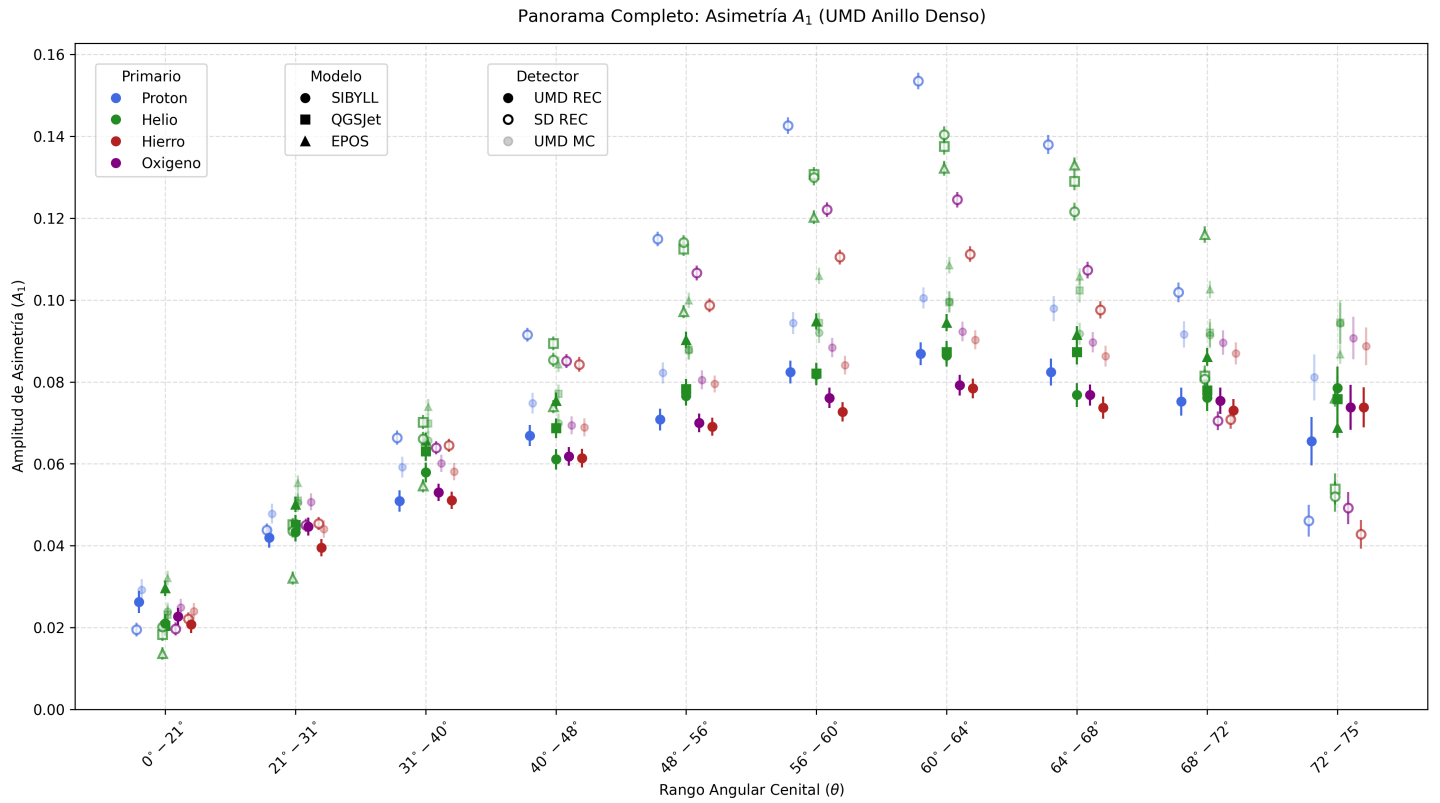


1. Panorama Global

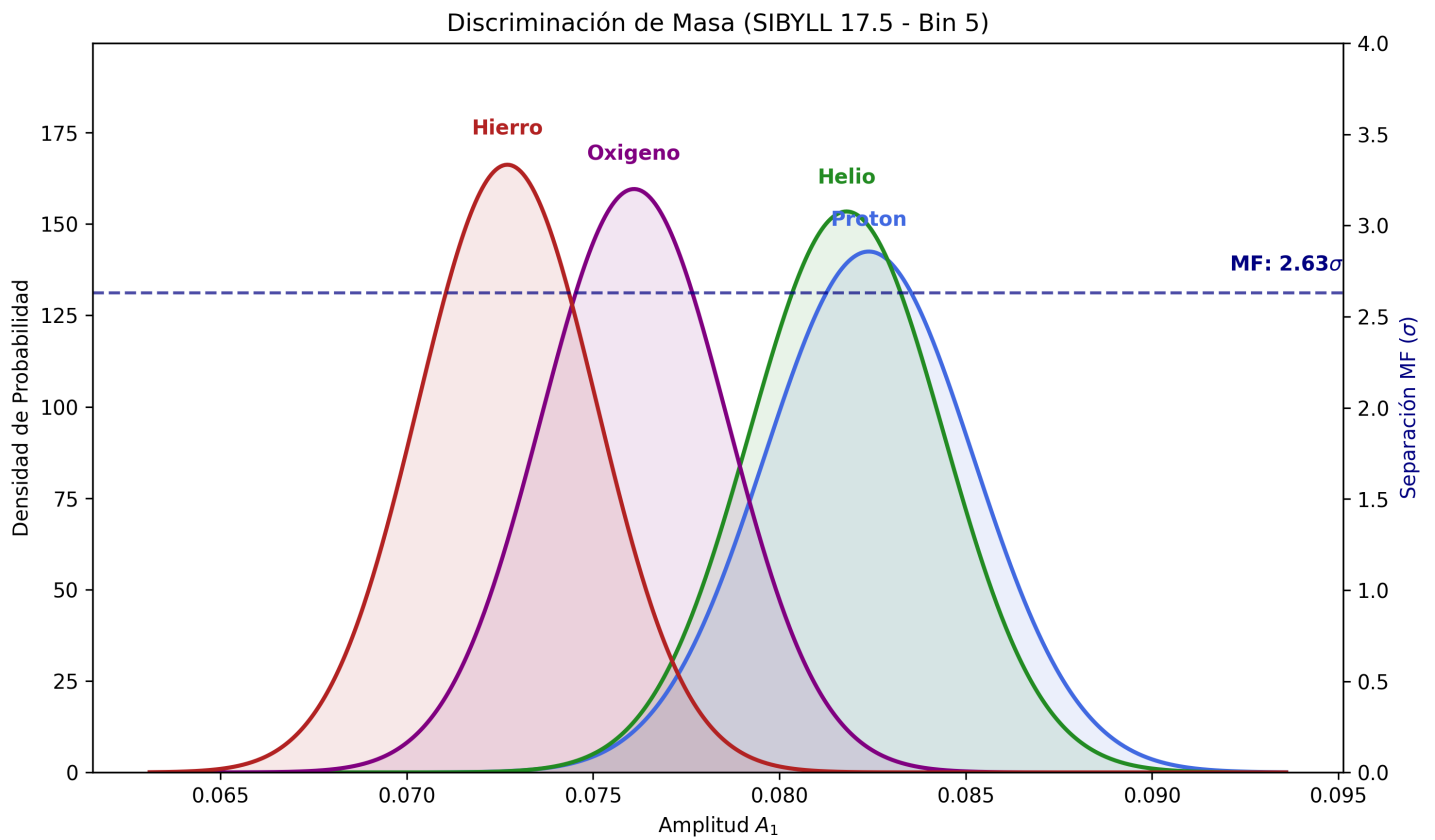


2. Discriminación de Masa (SIBYLL 17.5)

Se evalúa la capacidad de discriminación utilizando el Merit Factor (MF).

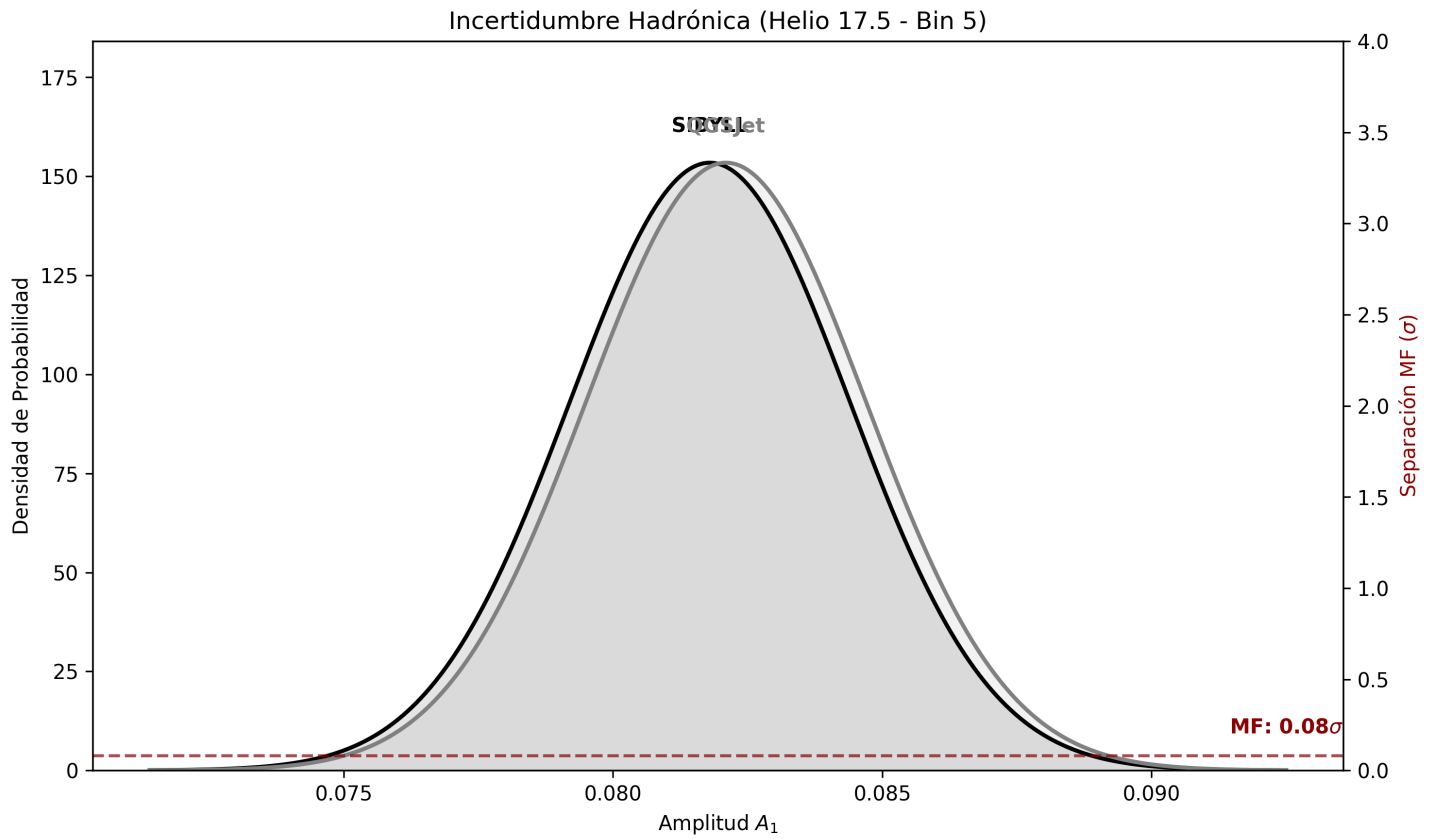
$$MF = |A1_a - A1_b| / \sqrt{\sigma_a^2 + \sigma_b^2}$$

Separación Protón vs Hierro observada: 2.63 sigma.

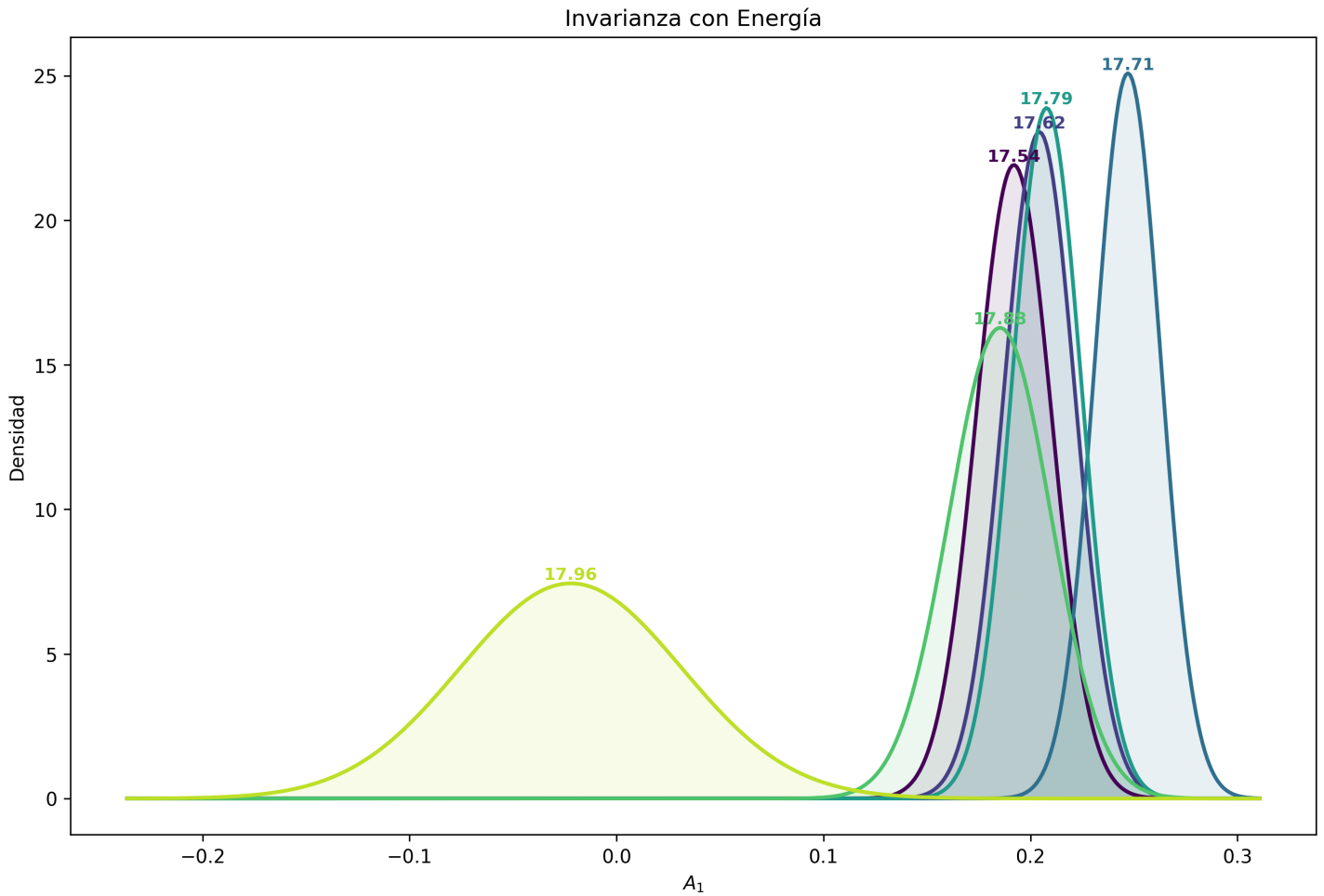


3. Modelos Hadrónicos (Helio 17.5)

Discrepancia entre modelos (QGS vs SIB): 0.08 sigma.



4. Sistemático: Energía



	17.54	17.62	17.71	17.79	17.88	17.96
17.54	-	0.5	2.3	0.6	0.2	3.8
17.62	0.5	-	1.8	0.1	0.6	4.0
17.71	2.3	1.8	-	1.7	2.1	4.8
17.79	0.6	0.1	1.7	-	0.8	4.1
17.88	0.2	0.6	2.1	0.8	-	3.5
17.96	3.8	4.0	4.8	4.1	3.5	-

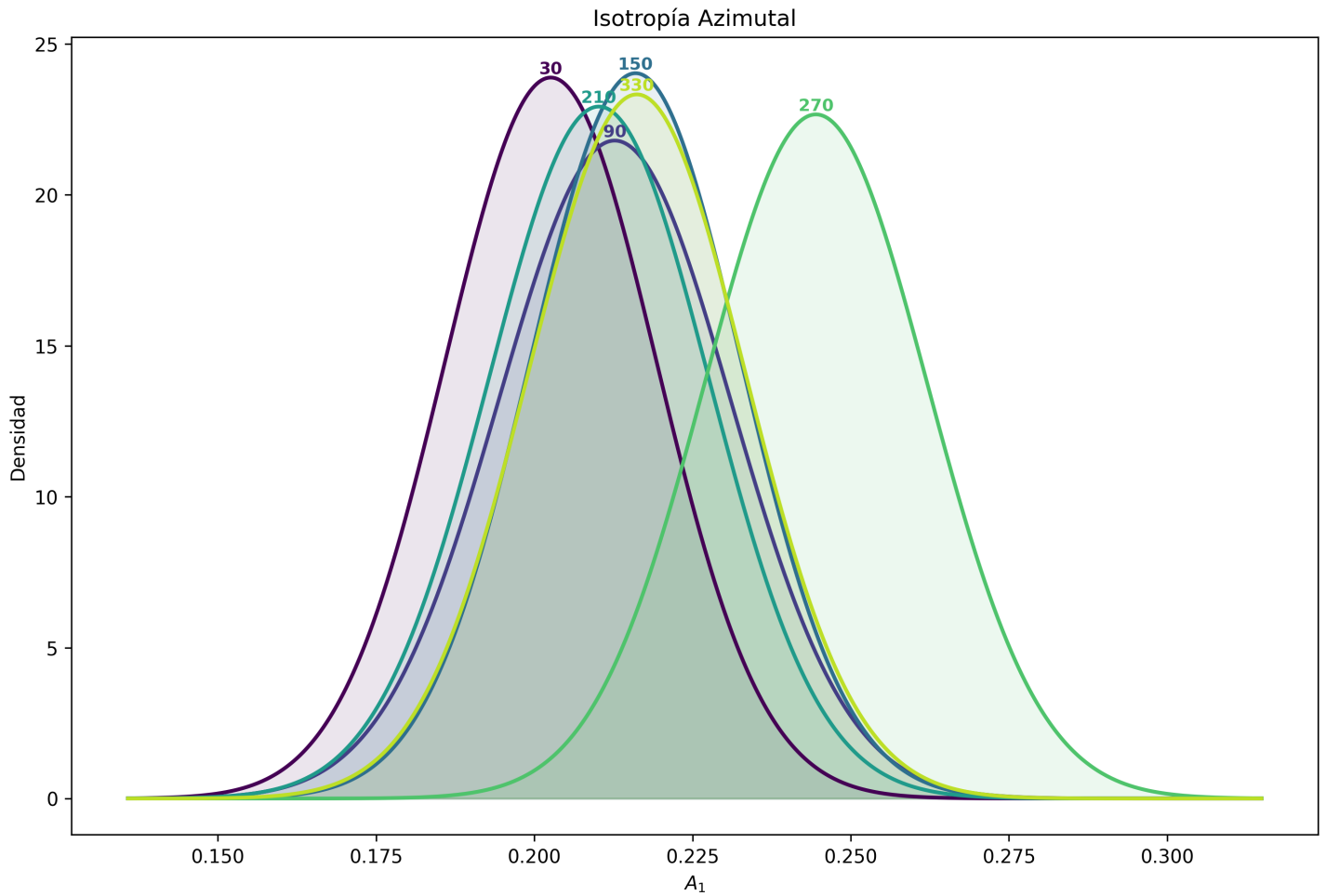
Matriz de Merit Factor (MF) entre energías:

E	17.54	17.62	17.71	17.79	17.88	17.96
17.54	-	0.49	2.28	0.64	0.22	3.78
17.62	0.49	-	1.82	0.15	0.63	4.02

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17.71	2.28	1.82	-	1.70	2.12	4.82
17.79	0.64	0.15	1.70	-	0.76	4.10
17.88	0.22	0.63	2.12	0.76	-	3.52
17.96	3.78	4.02	4.82	4.10	3.52	-

5. Sistemático: Isotropía



	30	90	150	210	270	330
30	-	0.4	0.6	0.3	1.7	0.6
90	0.4	-	0.1	0.1	1.3	0.1
150	0.6	0.1	-	0.2	1.2	0.0
210	0.3	0.1	0.2	-	1.4	0.2
270	1.7	1.3	1.2	1.4	-	1.2
330	0.6	0.1	0.0	0.2	1.2	-

Matriz de Merit Factor (MF):

Sec	30	90	150	210	270	330
30	-	0.41	0.57	0.32	1.73	0.57
90	0.41	-	0.13	0.10	1.25	0.14

Reporte Global de Tesis - Asimetría UMD/SD

150	0.57	0.13	-	0.24	1.18	0.01
210	0.32	0.10	0.24	-	1.39	0.25
270	1.73	1.25	1.18	1.39	-	1.15
330	0.57	0.14	0.01	0.25	1.15	-